Do We Train on Test Cells? Provenance Leakage in Public Cervical Cytology Benchmarks

Snehasish Ghosh*, Soham Mitra†, and Nibaran Das†
*Indian Institute of Science, Bengaluru, India
Email: snehasishg@iisc.ac.in
†Department of Computer Science and Engineering, Jadavpur University, Kolkata, India
Emails: [SOHAM EMAIL], [NIBARAN EMAIL]

Abstract—Cervical cancer caused an estimated 660,000 new cases and 350,000 deaths worldwide in 2022, making trustworthy automation for cytology screening a public-health priority [1]. Existing classifiers attain high crop-level accuracy on public cervical-cell benchmarks but often treat several cells from one microscopic field as independent observations, exposing train and test partitions to shared source fingerprints and, in some cases, identical pixels. We propose CytoFair, a provenance-recovery and grouped-evaluation framework that independently reconstructs crop-to-parent mappings, validates geometry, quantifies structural and literal overlap, tests parent identifiability, and releases fixed parent-disjoint splits. On SIPaKMeD, parent identification reaches $25.73 \pm 0.66\%$ versus $0.17 \pm 0.24\%$ for the label-shuffled comparator, while $92.62 \pm 0.40\%$ of random-split test crops have a training parent and $52.62 \pm 1.97\%$ share literal pixels; all six architecture-wise P1−P2 point estimates are positive, although none survives the preregistered Benjamini–Hochberg correction. Code, provenance maps, and fixed splits are available at [REPOSITORY URL].

Index Terms—cervical cytology, data leakage, provenance, grouped splitting, benchmark audit, reproducibility

I. Introduction

Cervical cytology is a high-volume screening task in which a small number of abnormal cells must be recognized among many visually plausible normal cells. The public-health burden is substantial: cervical cancer was the fourth leading cause of cancer incidence and mortality among women worldwide in 2022 [1]. Computer-assisted screening is therefore attractive not merely as a leaderboard problem, but as a way to support scarce expert attention. That promise depends on an elementary validity condition: the held-out sample should represent genuinely unseen biological or acquisition units. The image presented to a network, however, is not always the independent unit sampled from the population. When several released cell crops originate from one field, they inherit stain, illumination, focus, background texture, preparation artifacts, and sometimes overlapping pixels. A crop-level split can then put siblings on both sides of evaluation while appearing formally disjoint at the filename level.

Deep learning has substantially advanced cervical-cell analysis. DeepPap classified cells directly from images and reported 98.3% accuracy on Herlev [2]; DeepCervix fused features from multiple networks and reported 99.14% five-class accuracy on SIPaKMeD [3]; and a knowledge-distillation framework reported 98.52% on the same five-class task [4]. These methods draw on residual, dense, efficient, mobile, and transformer backbones whose general image-recognition capabilities are well established [5]–[9]. Public resources have made such comparisons possible: SIPaKMeD provides 4,049 cell crops derived from 966 cluster fields [10], Cx22 provides 1,320 generated regions with instance labels [11], CRIC provides 11,534 classified cells in 400 source images [12], and Herlev provides 917 single-cell images [13]. Yet performance engineering has advanced faster than auditing of the sampling hierarchy beneath these released observations.

Establishing defensible benchmark generalization is challenging for three concrete reasons. First, cervical classes can be morphologically similar, making small changes in test composition consequential. Second, public datasets are small and can be balanced by crop count while remaining highly imbalanced by source-parent count. Third, parent identifiers are not always carried into downstream crop folders, so ordinary stratified splitting silently discards the hierarchy that generated the data. The third difficulty is the focus of this work: we recover that hierarchy exactly where the release permits it, measure what crop splitting shares, and construct partitions that keep a parent—and, in a conservative sensitivity analysis, a near-duplicate component—wholly in one arm. We distinguish structural non-independence from causal classifier use: recoverable parent information shows that a shortcut is available, not that every class model necessarily uses it.

Our contributions are exactly fourfold:

- CytoFair, an auditable framework for exact crop-to-parent recovery, deterministic validation, and provenance-aware split generation;
- the first complete dual-branch provenance map for all 4,049 SIPaKMeD crops, together with source recovery for Cx22 and structural audits of CRIC and Herlev;
- direct measurements of random-split parent straddling, literal cross-split pixel sharing, and a five-seed parent-identity probe with transformation and shuffled-label controls;
- extensive experiments on SIPaKMeD, Cx22, CRIC,

and Herlev demonstrate that crop-level evaluation can be structurally non-independent and that fixed parent-disjoint splits remove the measured sharing.

II. Related Work

A. Cervical Cell Classification

Zhang et al. proposed DeepPap, achieving 98.3% accuracy on the Herlev benchmark; however, cervical crop-to-parent provenance was outside its scope [2]. Rahaman et al. proposed DeepCervix, achieving 99.14% five-class accuracy on SIPaKMeD; however, cervical crop-to-parent provenance was outside its scope [3]. Gao et al. proposed a transfer-learning and knowledge-distillation classifier, achieving 98.52% five-class accuracy on SIPaKMeD; however, cervical crop-to-parent provenance was outside its scope [4]. These numbers establish the maturity of crop classification, but they are not imported into our comparison because nominally identical datasets under different splits do not define comparable tests.

B. Leakage, Duplicates, and Identity

Barz and Denzler proposed ciFAIR, identifying training counterparts for 3.3% and 10% of CIFAR-10 and CIFAR-100 test images; however, cervical crop-to-parent provenance was outside its scope [14]. Laroca et al. proposed duplicate-free license-plate splits, re-evaluating six recognition models on AOLP and CCPD; however, cervical crop-to-parent provenance was outside its scope [15]. Packhäuser et al. proposed a chest-radiograph re-identification system, achieving 95.55% identity accuracy on ChestX-ray14; however, cervical crop-to-parent provenance was outside its scope [16]. Kapoor and Narayanan situated non-independence within a broader taxonomy of leakage and scientific reproducibility failures [17]. Table I makes the shared scope boundary explicit without labeling any individual study deficient. The gap remains an exact, validated provenance audit and corrected public evaluation resource for cervical-cytology crop benchmarks; this paper addresses it.

III. Method

Figure 1 summarizes the full pipeline. CytoFairtreats provenance recovery as a prerequisite, not a post-hoc clustering step: released or exactly reconstructed parent identity defines the independent grouping unit, while image-derived similarity is reserved for a separately labeled sensitivity protocol.

A. Exact Provenance Recovery

For each released crop x_i, let $g(x_i) = p_j$ denote its source parent. SIPaKMeD Branch A parses the released filename hierarchy, whereas Branch B ignores that assignment, searches all same-class parent fields, and selects the contained region with maximum normalized cross-correlation

$$\mathrm{NCC}(x, r) = \frac{\sum_u (x_u - \bar{x})(r_u - \bar{r})}{\sqrt{\sum_u (x_u - \bar{x})^2 \sum_u (r_u - \bar{r})^2}}. \tag{1}$$

Recovery is accepted only if the branches agree on at least 99% of common crops, every bounding box lies inside its parent, every crop has exactly one parent, crop totals equal the release, and the reconstructed class-parent counts equal the published vector. Cx22 parents and requested regions are decoded from the authors' MATLAB generator metadata and checked against the 686-image CCEDD source release; CRIC parent identities come from its released classifications table; one Herlev crop is defined as one parent.

B. Structural and Literal Sharing

For a parent with k crops assigned independently to 80% training and 20% test, the probability of straddling both arms is

$$P_{\mathrm{straddle}}(k) = 1 - 0.8^k - 0.2^k. \tag{2}$$

The analytic dataset expectation is the mean of (2) over parents. For an allocation with training-parent set G_{tr} and test crops T, we measure

$$L_p = \frac{|G_{\mathrm{tr}} \cap G_{\mathrm{te}}|}{|G|}, \quad L_c = \frac{1}{|T|} \sum_{x \in T} \mathbb{1}[g(x) \in G_{\mathrm{tr}}]. \tag{3}$$

For SIPaKMeD, the recovered bounding boxes additionally permit literal sharing: L_{pix} is the fraction of test crops whose parent has at least one training crop with nonempty source-coordinate intersection. Observed five-seed estimates are checked against an exact 10,000-allocation Monte Carlo interval under the implemented fixed-size, class-stratified rule.

C. Parent-Identity Mechanism Probe

Parents with at least four crops enter a balanced closed-set task; each eligible parent contributes the same number of training and test crops. A ResNet-18 f_θ is trained with cross-entropy over parent labels,

$$\mathcal{L}_{\mathrm{ID}}(\theta) = -\frac{1}{n} \sum_{i=1}^{n} \log \frac{\exp f_\theta(x_i)_{g(x_i)}}{\sum_p \exp f_\theta(x_i)_p}. \tag{4}$$

The mechanism test asks whether two different crops from one field retain a learnable common fingerprint after cellular content changes; a held-out accuracy of $25.73 \pm 0.66\%$ shows that the released crop alone contains strongly recoverable parent information. This establishes availability of a shortcut, not causal use by the downstream class classifier. The borrowed image transformations isolate candidate substrates: grayscale removes hue, channel-statistics matching follows Reinhard et al. [18], background-only masks use Otsu thresholds [19] followed by Telea inpainting [20], and high-pass filtering suppresses low-frequency illumination. The same images, optimizer, epochs, and seeds are retained for a parent-label-shuffled control.

TABLE I
Representative prior work and the common scope gap.

Work	Method / audit	Reported result	Limitations relative to this study
DeepPap [2]	Direct CNN	98.3% Herlev	Cervical parent provenance not studied
DeepCervix [3]	Feature fusion	99.14% SIPaKMeD	Cervical parent provenance not studied
Gao et al. [4]	Distillation	98.52% SIPaKMeD	Cervical parent provenance not studied
ciFAIR [14]	Duplicate audit	3.3/10% test duplicates	Cervical parent provenance not studied
Laroca et al. [15]	Fair LPR splits	Six models	Cervical parent provenance not studied
Packhäuser et al. [16]	Identity probe	95.55% ChestX-ray14	Cervical parent provenance not studied

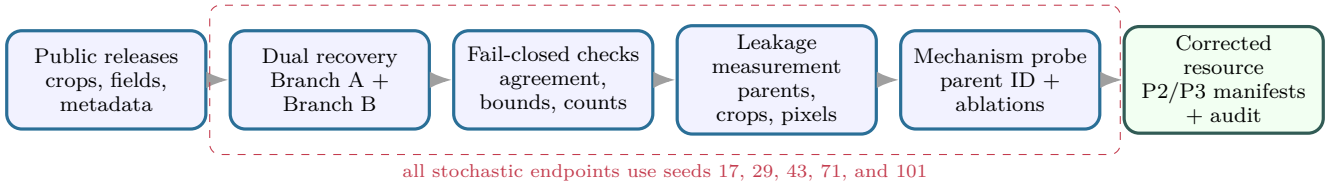


Fig. 1. The CytoFair pipeline. Exact provenance gates every downstream measurement; similarity-derived components affect only the conservative P3 sensitivity split.

Algorithm 1 Fail-closed provenance-aware evaluation

Require: Released observations X, source fields S, seeds $\mathcal{R}$

Recover $g_A : X \to S$ from released identifiers
Recover $g_B : X \to S$ by independent content matching
if agreement, containment, uniqueness, or count gates fail then
    stop and emit an audit ledger
for all $r \in \mathcal{R}$ do
    Build P1, P1′, P2, and P3
    Measure L_p, L_c, and, when boxes exist, L_{pix}
    Train the fixed probe and classifier matrix
Report mean ± sample SD, paired effects, and adjusted inference
Export provenance, split hashes, raw runs, and one-command Table-I build

D. Matched Evaluation Protocols

P1 is an 80:20 class-stratified crop split. P2 assigns complete recovered parents to one arm. P3 first joins parents connected by cross-parent candidates satisfying both 64-bit pHash and dHash Hamming thresholds of six, then groups the entire connected component; it is a sensitivity analysis, not a claim that every candidate is a proven duplicate. P1′ retains the P1 test set and, where required, downsamples only its training arm to the exact P2 training count. On SIPaKMeD the counts are already identical, so P1′=P1; both preregistered contrasts remain in the multiplicity family for protocol fidelity and are not independent corroboration.

E. Backbones and Confirmatory Statistics

The borrowed classifier component comprises ImageNet-pretrained ResNet-18/50 [5], DenseNet-121 [6], MobileNet-V2 [7], EfficientNet-B0 [8], and ViT-B/16 [9]; only their final heads are replaced. For each architecture we pair identical seeds across protocols. Normal paired differences receive a two-sided paired t-test and Cohen's d_z; otherwise the preregistered fallback is Wilcoxon with rank-biserial correlation. All P1−P2 and P1′−P2 classifier tests form one 12-test Benjamini–Hochberg family [21]; effect-size intervals use 10,000 paired bootstrap resamples.

TABLE II
SIPaKMeD counts and P1 seed-17 split. Counts are deterministic.

Class	Crops	Parents	Train	Test
Superficial–Intermediate	831	126	665	166
Parabasal	787	108	630	157
Koilocytotic	825	238	660	165
Metaplastic	793	271	634	159
Dyskeratotic	813	223	650	163
Total	4,049	966	3,239	810

IV. Experiments

A. Datasets and Integrity Checks

Table II gives the independently reconstructed SIPaKMeD class counts and one representative class-stratified P1 allocation. The 4,049 crops map to 966 parents, Cx22's 1,320 generated regions map to 514 recovered CCEDD sources, CRIC's 11,534 labeled cells map to 400 images, and Herlev contains 917 one-crop parents [10]–[13]. Branch A and Branch B agree on 4,049/4,049 SIPaKMeD crops, all recovered boxes are contained, and the class-parent vector is 126/108/238/271/223. No observation is excluded.

B. Implementation Details

Every stochastic cell uses seeds 17, 29, 43, 71, and 101 and is reported as mean ± sample SD. Table III records the complete training configuration. Training used deterministic CUDA kernels where available and at most three NVIDIA RTX A6000 GPUs concurrently; each individual run used one GPU.

TABLE III
Fixed hyperparameters for the frozen experiment.

Setting	Cell classifier	Parent probe
Optimizer	AdamW	AdamW
Learning rate	10^{-4}	10^{-4}
Weight decay	10^{-4}	10^{-4}
Batch size	64	64
Epochs	20	15
Input	224×224 RGB	224×224
Augmentation	resized crop, H/V flip	resized crop, H/V flip
Initialization	ImageNet pretrained	ImageNet pretrained
Seeds	17, 29, 43, 71, 101	
Hardware	NVIDIA RTX A6000; $\leq$3 concurrent GPUs	
Software	PyTorch 2.9.0, torchvision 0.24.0, CUDA 12.8	

C. Metrics

For N test crops and C classes, with true label y_i and prediction $\hat{y}_i$, we report

$$\mathrm{Acc} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[y_i = \hat{y}_i], \tag{5}$$

$$\mathrm{MacroF1} = \frac{1}{C}\sum_{c=1}^{C} \frac{2P_cR_c}{P_c + R_c}, \tag{6}$$

$$\mathrm{BAcc} = \frac{1}{C}\sum_{c=1}^{C} \frac{TP_c}{TP_c + FN_c}. \tag{7}$$

Accuracy is the preregistered primary endpoint; macro-F1 and balanced accuracy are retained concordance metrics. Structural endpoints are L_p, L_c, and L_{pix} from (3).

D. Main Comparison

Table IV is the central comparison. The observed straddle means fall inside their exact 10,000-allocation 95% intervals for all four datasets. SIPaKMeD has $45.63 \pm 1.04\%$ straddled parents and $92.62 \pm 0.40\%$ of test crops with a training parent; Cx22 has $34.01 \pm 1.10\%$ and $85.23 \pm 3.51\%$; CRIC reaches $94.35 \pm 1.10\%$ and $99.97 \pm 0.08\%$. Herlev, the structural negative control, remains exactly $0.00 \pm 0.00\%$. Both grouped protocols eliminate measured parent sharing. Among 13,064 within-parent SIPaKMeD crop pairs, 2,185 overlap; under P1, $52.62 \pm 1.97\%$ of test crops share literal source pixels with a training crop, versus $0.00 \pm 0.00\%$ under P2/P3.

Table V compares all six architectures under identical training settings and split sizes; the corrected CytoFair P2 column is bold. Every P1$-$P2 point estimate is positive, ranging from 0.72 ± 0.83 to 2.25 ± 1.63 percentage points, but protocol sensitivity is not monotone for every model.

E. Mechanism Ablation

The probe ablation in Table VI shows that parent information persists after every prespecified transformation. Background-only accuracy is $18.68 \pm 1.25\%$, so the signal is not confined to cellular morphology; persistence elsewhere does not identify a single exclusive substrate. The label-shuffled run is $0.17 \pm 0.24\%$, with a five-seed 95% interval of -0.14% to 0.47% that covers the exact 0.2358% chance baseline.

F. Significance and Registered Predictions

Table VII reports every primary P1$-$P2 test with its paired effect. The identical six P1$'-$P2 rows are retained in the registered 12-test family, rather than presented as independent evidence. No contrast survives correction; the smallest adjusted value is $q = 0.1109$. Accordingly, the classifier experiment estimates direction and scale of protocol sensitivity, not a confirmatory universal inflation effect. Architecture ordering also changes: Kendall's $\tau = 0.467$ (seed-bootstrap 95% CI 0.071–0.733, unadjusted $p = 0.2722$), with 4/15 pairwise reversals.

The registered monotonic accuracy hypothesis is not supported: P3 exceeds P2 in four of six architecture means (Table VIII). This failure is retained rather than repaired. Because P3 changes grouped test composition, stronger structural conservatism does not imply monotonically lower finite-sample accuracy.

The frozen run schema retained aggregate accuracy, macro-F1, balanced accuracy, and epoch histories but not final per-sample predictions or checkpoints. A post-hoc class confusion matrix would therefore require new inference after the declared hard stop; we do not fabricate one or infer class-pair errors from aggregate metrics. Figure 2 supplies the qualitative provenance view and the retained loss trajectory.

V. Discussion

The main finding is structural rather than architectural. Random crop splitting makes the nominal test set extensively non-independent: most SIPaKMeD test crops have a training parent, more than half share literal source pixels with a training crop, and a model can identify the parent far above chance even from background-only inputs. The same measurement correctly returns zero for the one-cell-per-parent Herlev structural negative control, while the high-crop-per-field CRIC construction produces the largest straddle rate. These construction-linked results, dual-branch agreement, and analytic sanity check are mutually independent safeguards against a split-implementation artifact.

The two weakest architecture–protocol cells are MobileNet-V2 under P2 ($95.16 \pm 1.50\%$) and ViT-B/16 under P3 ($95.53 \pm 1.68\%$). Their interpretation must remain aggregate because a class confusion matrix is unavailable in the frozen evidence. Two real dataset properties nevertheless explain why grouped test composition can be unstable without claiming a particular error pair: crop counts are nearly balanced (787–831 per class), whereas parent counts range from 108 Parabasal fields to 271 Metaplastic fields, and cervical categories are defined by related morphological characteristics rather than separated acquisition domains. Thus a parent-disjoint fold can preserve crop totals while changing the diversity of independently sourced appearances. The larger ViT-B/16 P3 SD is consistent with such composition sensitivity, but is not proof of a class-specific mechanism.

TABLE IV
Structural leakage under crop-level and grouped protocols. Stochastic entries are mean ± sample SD over five fixed seeds.

Dataset	Crops	Parents	Analytic P1 straddle	Observed P1 straddle	P1 test crop with train parent	P2 straddle	P3 straddle
SIPaKMeD	4,049	966	44.83%	45.63 ± 1.04%	92.62 ± 0.40%	0.00 ± 0.00%	0.00 ± 0.00%
Cx22	1,320	514	33.93%	34.01 ± 1.10%	85.23 ± 3.51%	0.00 ± 0.00%	0.00 ± 0.00%
CRIC	11,534	400	94.88%	94.35 ± 1.10%	99.97 ± 0.08%	0.00 ± 0.00%	0.00 ± 0.00%
Herlev	917	917	0.00%	0.00 ± 0.00%	0.00 ± 0.00%	0.00 ± 0.00%	0.00 ± 0.00%

TABLE V
SIPaKMeD test accuracy. All models were rerun in this study under the same 3,239/810 train/test counts. Mean ± sample SD over five seeds.

Architecture	P1 crop	P1′ size-matched	CytoFair P2 parent-disjoint	P3 component-disjoint
DenseNet-121	98.12 ± 0.28%	98.12 ± 0.28%	96.72 ± 0.69%	97.38 ± 0.50%
EfficientNet-B0	97.78 ± 0.49%	97.78 ± 0.49%	96.69 ± 1.49%	97.41 ± 0.71%
MobileNet-V2	97.41 ± 0.45%	97.41 ± 0.45%	95.16 ± 1.50%	96.20 ± 0.93%
ResNet-18	97.19 ± 0.59%	97.19 ± 0.59%	96.00 ± 0.89%	96.35 ± 1.28%
ResNet-50	97.51 ± 0.47%	97.51 ± 0.47%	96.79 ± 0.42%	96.27 ± 0.53%
ViT-B/16	97.14 ± 0.95%	97.14 ± 0.95%	95.56 ± 0.84%	95.53 ± 1.68%

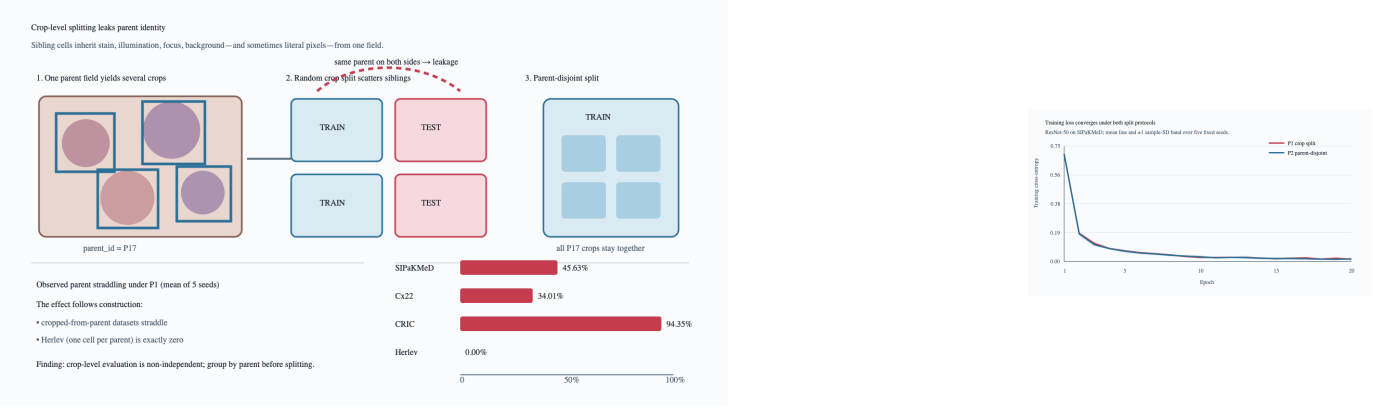


Fig. 2. Qualitative provenance and optimization diagnostics. Left: sibling crops can inherit a common field fingerprint, while parent grouping keeps them together. Right: frozen ResNet-50 training-loss histories converge under both split protocols.

TABLE VI
Parent-ID probe ablation on the balanced 424-way task.

Input condition	Accuracy	Multiple of chance
RGB	25.73 ± 0.66%	109.1×
Grayscale	13.80 ± 1.12%	58.5×
Stain-normalized	17.31 ± 0.64%	73.4×
Background only	18.68 ± 1.25%	79.2×
High-pass	16.27 ± 0.60%	69.0×
Shuffled labels	0.17 ± 0.24%	0.7×

All six P1−P2 estimates are positive, yet none survives the declared BH family, and four architectures violate the registered P3≤P2≤P1 prediction. We therefore do not claim universal accuracy inflation or causal shortcut use. The justified conclusion is narrower: the crop is not the independent sampling unit, the source signal is available to a learner, and P2 removes the measured parent and pixel sharing while preserving the same SIPaKMeD crop counts.

VI. Limitations

The audited public datasets are modest in independent-parent count, and only SIPaKMeD received the complete six-architecture rescore, limiting claims about the magnitude of downstream performance changes. No prospective clinical validation, patient-level outcome study, or deployment analysis was performed, so the corrected benchmark scores must not be read as estimates of clinical utility. Training was constrained to at most three RTX A6000 GPUs, which limited broader full-matrix reruns, and the frozen aggregate run artifacts cannot support a post-hoc class confusion matrix without violating the hard stop.

VII. Conclusion

We investigated whether public cervical-cytology classifiers were evaluated on genuinely independent cells and found that crop filenames concealed a consequential source-field hierarchy. CytoFair recovered exact provenance, validated the reconstruction independently, measured parent straddling and literal pixel sharing, demonstrated a learnable parent-field fingerprint, and produced fixed parent-disjoint splits. On SIPaKMeD, most crop-level test observations had a parent in training and more than half shared source pixels with a training crop; Cx22 and CRIC showed the same structural pattern, while Herlev behaved as the expected zero-leakage control. Across six classifiers, ordinary crop splitting yielded positive P1−P2 point estimates for every architecture, but the preregistered multiplicity-corrected evidence did not establish a universal accuracy-inflation effect. The failed monotonic P3 prediction was retained and narrowed the claim further: provenance-aware grouping corrects the

TABLE VII
Preregistered paired P1−P2 inference. Δ is percentage points, d_z is paired Cohen's effect, and q is BH-adjusted over the declared 12-test family.

Architecture	Δ, mean $\pm$ SD	Raw p	BH q	Cohen's d_z	Bootstrap 95% CI for d_z
DenseNet-121	1.41 ± 0.90	0.0251	0.1109	1.561	[0.985, 4.747]
EfficientNet-B0	1.09 ± 1.85	0.2591	0.2591	0.588	[−0.316, 4.668]
MobileNet-V2	2.25 ± 1.63	0.0370	0.1109	1.377	[0.809, 6.686]
ResNet-18	1.19 ± 1.28	0.1081	0.1511	0.922	[0.442, 2.365]
ResNet-50	0.72 ± 0.83	0.1260	0.1511	0.863	[0.085, 4.025]
ViT-B/16	1.58 ± 1.70	0.1059	0.1511	0.931	[0.057, 4.950]

TABLE VIII
Registered protocol-ordering diagnostic. This is not a class confusion matrix.

Architecture	P3≤P2≤P1
DenseNet-121	Not supported
EfficientNet-B0	Not supported
MobileNet-V2	Not supported
ResNet-18	Not supported
ResNet-50	Supported
ViT-B/16	Supported

evaluation unit, but a more conservative group definition need not monotonically reduce finite-sample accuracy. The resulting contribution was therefore a benchmark correction and reusable measurement resource rather than another incremental classifier. In V2, we will preregister a provenance-aware instance-segmentation evaluation on Cx22 and extend the same grouping apparatus to specimen-disjoint multi-view Fish-Pak and temporally grouped Croatian Fish video, with deterministic manifests fixed before model training.

References

[1] World Health Organization, "Human papillomavirus and cancer," WHO Fact Sheet, 2025, accessed: 2026-08-22. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/human-papilloma-virus-and-cancer

[2] L. Zhang, L. Lu, I. Nogues, R. M. Summers, S. Liu, and J. Yao, "DeepPap: Deep convolutional networks for cervical cell classification," IEEE Journal of Biomedical and Health Informatics, vol. 21, no. 6, pp. 1633–1643, 2017.

[3] M. M. Rahaman et al., "DeepCervix: A deep learning-based framework for the classification of cervical cells using hybrid deep feature fusion techniques," Computers in Biology and Medicine, vol. 136, p. 104649, 2021.

[4] W. Gao, C. Xu, G. Li, Y. Zhang, N. Bai, and M. Li, "Cervical cell image classification-based knowledge distillation," Biomimetics, vol. 7, no. 4, p. 195, 2022.

[5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.

[6] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely connected convolutional networks," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2017, pp. 4700–4708.

[7] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR), 2018, pp. 4510–4520.

[8] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in Proc. 36th Int. Conf. Machine Learning, ser. Proceedings of Machine Learning Research, vol. 97. PMLR, 2019, pp. 6105–6114.

[9] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in Proc. Int. Conf. Learning Representations (ICLR), 2021.

[10] M. E. Plissiti, P. Dimitrakopoulos, G. Sfikas, C. Nikou, O. Krikoni, and A. Charchanti, "SIPaKMeD: A new dataset for feature and image based classification of normal and pathological cervical cells in Pap smear images," in Proc. IEEE Int. Conf. Image Process. (ICIP), 2018, pp. 3144–3148.

[11] G. Liu, Q. Ding, H. Luo, M. Sha, X. Li, and M. Ju, "Cx22: A new publicly available dataset for deep learning-based segmentation of cervical cytology images," Computers in Biology and Medicine, vol. 150, p. 106194, 2022.

[12] M. T. Rezende et al., "CRIC searchable image database as a public platform for conventional Pap smear cytology data," Scientific Data, vol. 8, no. 1, p. 151, 2021.

[13] J. Jantzen, J. Norup, G. Dounias, and B. Bjerregaard, "Pap-smear benchmark data for pattern classification," in Proc. Nature Inspired Smart Information Systems (NiSIS). Albufeira, Portugal: NiSIS, 2005, pp. 1–9.

[14] B. Barz and J. Denzler, "Do we train on test data? purging CIFAR of near-duplicates," Journal of Imaging, vol. 6, no. 6, p. 41, 2020.

[15] R. Laroca, V. Estevam, A. S. B. Jr., R. Minetto, and D. Menotti, "Do we train on test data? the impact of near-duplicates on license plate recognition," in Proc. Int. Joint Conf. Neural Networks (IJCNN), 2023, pp. 1–8.

[16] K. Packhäuser, S. Gündel, N. Münster, C. Syben, V. Christlein, and A. Maier, "Deep learning-based patient re-identification is able to exploit the biometric nature of medical chest X-ray data," Scientific Reports, vol. 12, p. 14851, 2022.

[17] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, p. 100804, 2023.

[18] E. Reinhard, M. Ashikhmin, B. Gooch, and P. Shirley, "Color transfer between images," IEEE Computer Graphics and Applications, vol. 21, no. 5, pp. 34–41, 2001.

[19] N. Otsu, "A threshold selection method from gray-level histograms," IEEE Transactions on Systems, Man, and Cybernetics, vol. 9, no. 1, pp. 62–66, 1979.

[20] A. Telea, "An image inpainting technique based on the fast marching method," Journal of Graphics Tools, vol. 9, no. 1, pp. 25–36, 2004.

[21] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: A practical and powerful approach to multiple testing," Journal of the Royal Statistical Society: Series B (Methodological), vol. 57, no. 1, pp. 289–300, 1995.